



Estimation and Sampling of Unnormalized Statistical Models with Stein Score Matching



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Abstract

Stein's method provides a powerful tool for handling probabilistic distributions with differential operators. Recently there has been a surge of research interest in kernelizing Stein methods, covering a wide range of important applications such as Bayesian inference, generative modeling, and goodness-of-fit tests, among many others. This study intends to fill a gap in the kernelized Stein literature, by addressing the problem of estimation and sampling of unnormalized statistical models based on empirical examples, a challenge shared by both the machine learning and statistics community. Experimental evidence shows encouraging results.

Parameter Estimation for Unnormalized Models

Notations: We consider P as the target distribution and Q as the model distribution. We denote $s_p = \nabla_x \log p(x)$ as the score function.

Existing Methods:

- Hyvärinen's Score Matching (SM) [Hyvärinen, 2005]

$$\min_Q \mathcal{D}(P, Q) = \frac{1}{2} \int p(x) \|s_p(x) - s_q(x)\|^2 dx$$

$$= \mathbb{E}_{\{x \sim P\}} \left[\Delta_x \log q(x) + \frac{1}{2} \|\nabla_x \log q(x)\|^2 \right] + C$$

- Only requires the data samples. 😊
- Costly computation from second-order Laplacian. 😞

- Noise Contrastive Estimation (NCE) [Gutmann, 2010]

$$\max_{\{\theta, c\}} \mathcal{J}_{\text{NCE}}(\theta, c) = \frac{1}{2n} \sum_i (\log h(x_i; \theta) + \log(1 - h(x'_i; \theta)))$$

where $h(x; \theta) = \sigma(r(x; \theta))$; $r(x; \theta) = \log \tilde{p}_\theta(x) - \log t(x) - c$
 $x \sim \tilde{p}_\theta(x) = p_\theta(x) / \exp(c)$ and $x' \sim t(x')$ is some noise distribution.

- It is Fisher efficient. 😊
- Requires a tractable noise distribution. The performance is very sensitive to the noise distribution. 😞

Stein's Score Matching (SSM) [Our method]

- Kernelized Stein's Discrepancy (KSD) [Liu, 2016]

$$\mathcal{S}(P, Q) = \mathbb{E}_{\{x, x' \sim P\}} [u_q(x, x')]$$

where $u_q(x, x') = s_q(x)^T k(x, x') s_q(x') + s_q(x)^T \nabla_{x'} k(x, x') + \nabla_x k(x, x')^T s_q(x') + \text{trace}(\nabla_{x, x'} k(x, x'))$

- Only requires the data samples and using first-order gradients. 😊

Result on four unnormalized distributions:

- Kidney: $\frac{1}{2} \left(\frac{\|x - \mu_1\|_{\sigma_1}^2}{\sigma_1^2} \right) - \log \left(e^{-\frac{1}{2} \frac{(x_1 - \mu_2)^2}{\sigma_2^2}} + e^{-\frac{1}{2} \frac{(x_1 + \mu_2)^2}{\sigma_2^2}} \right)$
- River: $-\ln \left(e^{-\frac{1}{2} \frac{x_2 - w_1(x; \sigma_3)^2}{\sigma_1^2}} + e^{-\frac{1}{2} \frac{x_2 - w_1(x; \sigma_3) + w_3(x; \sigma_4, \mu_1)^2}{\sigma_2^2}} \right)$
- Banana: $\frac{1}{2} \left[\left(\frac{x_1 - (x_2/\kappa)^2}{\sigma_1} \right)^2 + \left(\frac{x_2 - \mu_2}{\sigma_2} \right)^2 \right]$
- Wave: $\frac{1}{2} \left[\frac{x_2 - w_1(x; \sigma_2)}{\sigma_1} \right]^2$

Table 1: MSE Comparison for Parameter Estimation (Scaled to $1e-3$)

Dataset	Kidney	River	Banana	Wave
SSM (Ours)	6.926 ± .525	1.662 ± .268	2.681 ± .563	0.436 ± .085
SM [2]	5.011 ± .455	1.759 ± .264	2.695 ± .532	0.328 ± .078

Table 2: Runtime Comparison for Parameter Estimation (Seconds)

Dataset	Kidney	River	Banana	Wave
SSM (Ours)	0.82 ± .04	1.02 ± .03	10.90 ± 1.19	3.91 ± .07
SM [2]	1.45 ± .05	1.85 ± .05	10.21 ± 1.14	1.35 ± .02

Sampling for Unnormalized Models

- Langevin Itô Diffusion Equation:

$$dX_t = \nabla \log p(X_t) + \sqrt{2} dW_t$$
- We use a neural network to learn $s_p(x)$ by minimizing KSD.

$$\mathcal{L} = \mathcal{S}(P, Q) = \mathbb{E}_{\{x, x' \sim P\}} [u_q(x, x')]$$
- The learned score function can be used as the drift term $\nabla \log p(X_t)$ of Langevin flow to generate samples.

Experiments:

- Experiment 1: Learn the score function of a 2D Gaussian .

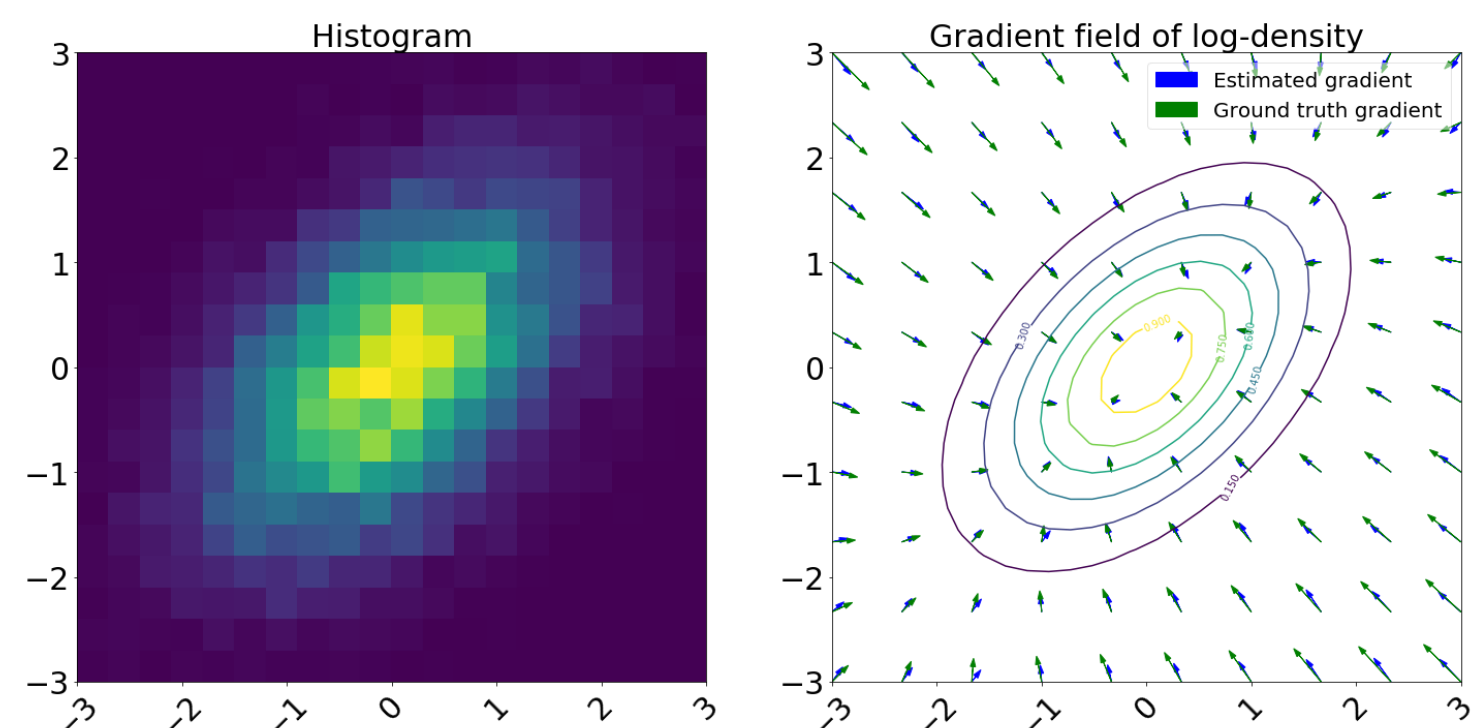


Figure 1. (Left) 2D Gaussian distribution; (Right) The gradient vector field. Blue: estimated gradient using KSD, Green: ground truth gradient.

- Experiment 2: Generate unseen data based on the MNIST dataset. We project the data onto a low-dimensional latent space and learn the score function using MLP. Next we sample from the latent space using Langevin diffusion and the learned score function. Lastly, we convert samples in the latent space into real images.

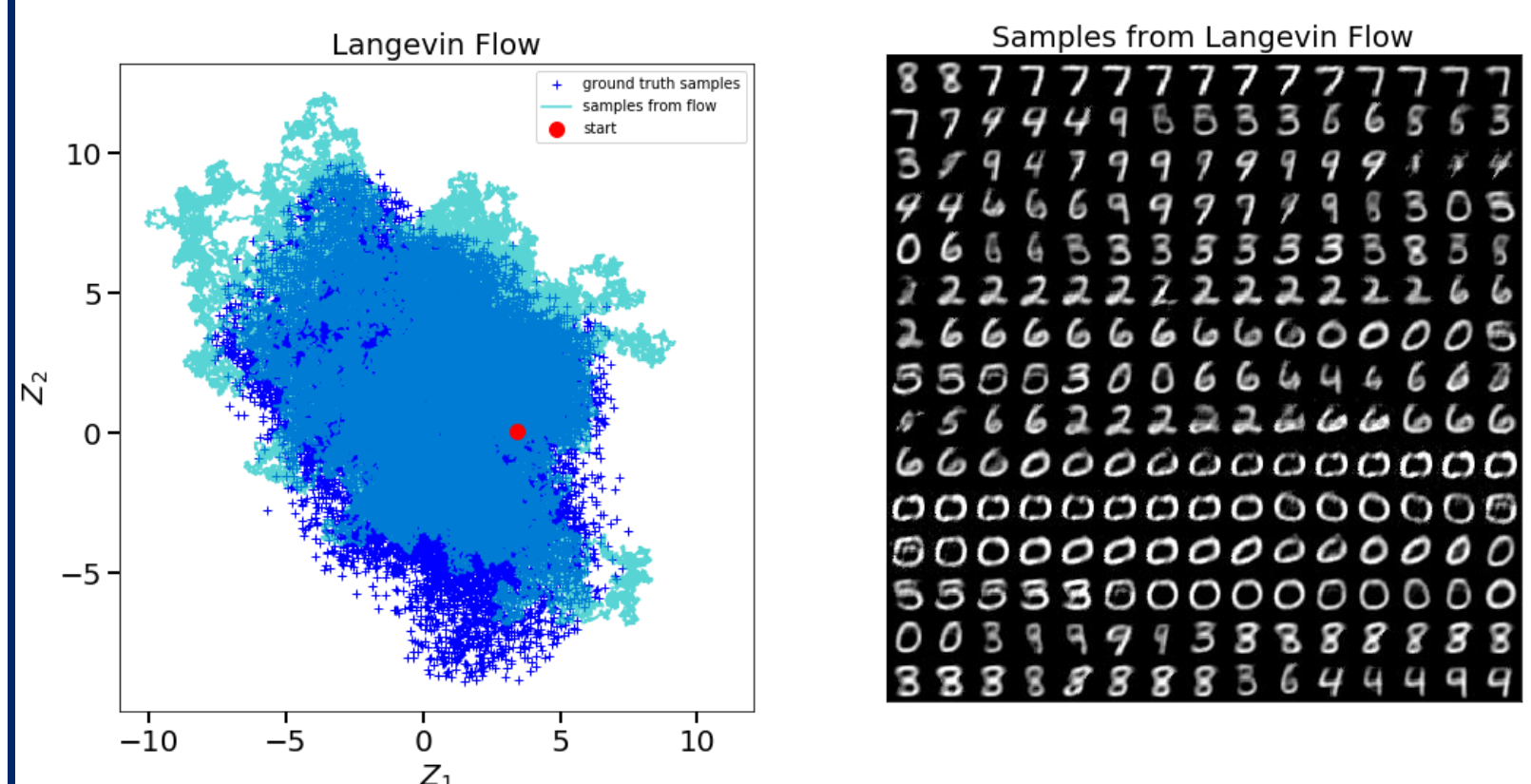


Figure 2. (Left) Langevin flow using the score as the drift term in the feature space; (Right) Generated samples.

Conclusion

Contributions:

- Demonstrate that SSM can produce competitive results, matching the ground truth.
- Propose a *novel* generative model by combining SSM and Langevin diffusion given samples from the target distribution.

Future Work

- Extend SSM to high-dimensional datasets and explore alternative formulation of Stein's discrepancy which avoids kernels.

References

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